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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

BIOSTATISTICS

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Explain :

   (a) Primary data.

   (b) Secondary data, and

   (c) Census method of data collection.

2. In a town with a population of 50,000, there are 2,000 births and 200 infant deaths in the year 1980. Of these 85 infants died in the first 28 days of life and 35 of them died in the first week of life. There were 110 still births in the same year. Calculate infant mortality rate, pre-natal, neonatal and post neonatal mortality rates

3. Calculate the Arithmetic Mean, Harmonic Mean, Median from the following data :

30, 40, 50, 10, 20, 30.

4. Find the quartiles and coefficient of quartile deviation from the following data :

2, 22, 4, 6, 8, 10, 12, 14, 16, 18, 20.

5. Find the regression coefficient of (a) x on y and (b) y on x when the correlation coefficient is + 0.8, standard deviation of 'x' is 3.5 and standard deviation of 'y' is 2.5.

6. What is the probability that a leap year, selected at random, will have 53 Sundays ?

7. The mean of a binomial distribution is 20 and standard deviation is 4. Find 'n', 'p', 'q'.

8. Explain inpatient bed occupancy ratio.

Part - B $(4 \times 10 = 40)$

Answer any **four** questions.

9. The following are the marks obtained by 40 students.

41 17 33 63 54 92 60 58 70 06 33 44 57 49 34 74
54 63 36 52

60 33 09 79 28 30 42 93 43 80 57 67 24 64 63 11
35 82 10 23

Construct a Frequency Table with class interval 10 and calculate the Arithmetic mean.

10. Compute coefficient of variance from the following data.

Class	31–35	36–40	41–45	46–50	51–55	56–60	61–65	66–70
Frequency	2	3	8	12	16	5	2	2

11. For 5 patients, temperature and pulse are given. Find the Correlation Coefficient.

Temperature	102	101	100	99	98
Pulse	100	90	80	70	60

12. Three groups of children contain 3 girls and 1 boy, 2 girls and 2 boys, and 1 girl and 3 boys respectively. One child is selected at random from each group. Find the probability that the three children selected include 1 girl and 2 boys.

13. Find both regression equations from the following data by taking deviation form actual mean of x and y.

x	10	12	13	12	16	15
y	40	38	43	45	37	43

14. The scores made by candidate in a certain test are normally distributed with mean 500 and deviation 100. What percentage of candidates receive scores

(a) less than 400

(b) between 400 and 500 ?

(Given $P(0 \leq Z \leq 1) = 0.34134$.

15. A certain drug was administered to 500 people out of a total of 800 included in the sample to test its efficacy against typhoid. The results are given below :

	Typhoid	No Typhoid	Total
Drug	200	300	500
No Drug	280	20	300
Total	480	320	800

On the basis of these data, can it be concluded that the drug is effective in preventing typhoid ? (Given for 1 degrees of freedom $\chi^2_{0.05} = 3.84$)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

GENERAL ANATOMY AND PHYSIOLOGY

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Section - A (5 × 6 = 30)

Answer any **five** questions.

1. Anatomy of small intestine.

2. Draw the structure of ear and name its parts.

3. Draw and label a neat diagram of heart.

4. Write short notes on pituitary gland.

5. Draw and label a neat diagram of urinary system.

6. Draw and label a neat diagram of digestive system.

7. Describe grossly structure of artery, arteriole and capillaries with the help of diagram.

8. Draw and lable neat diagram of eye.

Section - B $(4 \times 10 = 40)$

Answer any **four** questions.

9. Draw and explain the female reproductive system.

10. Explain about central nervous system.

11. Explain about muscular system and its functions.

12. Discuss about circulatory system.

13. Write clear notes about air exchange in human body. Which artery brings impure blood ?

14. What are all the endocrine glands you know ? Write few lines about each gland. Includes functions of hypo and hyper secretions.

15. Draw and label the picture of pancreas and write its functions.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

GENERAL AND OCULAR BIOCHEMISTRY

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Water soluble vitamin.

2. Polysaccharides.

3. Amino acids.

4. Biochemistry of vitreous.

5. Total count/Diffential count.

6. Biochemistry of tear film.

7. Blood grouping.

8. Enzyme.

Part - B (4 × 10 = 40)

Answer any **four** questions.

9. Glycolysis.

10. Properties of protein.

11. Biochemistry of lens.

12. Discuss ketone bodies.

13. Fatty acid biosynthesis.

14. Fat soluble vitamins.

15. Mechanism of enzyme action.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

GEOMETRICAL OPTICS

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Write short notes on laws of reflection.

2. Explain about dual nature of light.

3. Describe Fermat's principle.

4. Explain about reflecting prisms.

5. Explain about the geometrical theory of optical fibres.

6. Explain about different types of aberrations and their effects.

7. Explain about prisms.

8. Explain about dioptric power of the lens.

Part - B $(4 \times 10 = 40)$

Answer any **four** questions.

9. Derive the law's of refraction based on Fermat's principle.

10. Describe Snell's laws of refraction.

11. Describe in detail about refraction at convex surface.

12. Explain in detail about reflecting prisms.

13. Derive the relations of vergence equation.

14. Prove the relation between U, V and R for the case of convex lens.

15. Describe the geometrical theory of optical fibres.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

ENGLISH AND COMMUNICATION SKILLS

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Explain :

   (a) Statement.

   (b) Imperative sentences.

2. Compare Phrase and Clause.

3. Write a paragraph on Eve teasing.

4. What are the hits that to be followed in writing a story ?

5. Write a letter to your friend about your recent tour to Delhi.

6. List the features of effective speech.

7. What is persuasive skill ? –Explain.

8. Why should one be very strict with time during his presentation ?

Part - B (4 × 10 = 40)

Answer any **four** questions.

9. Write an essay on 'Parts of Speech'. Give suitable examples.

10. How are sentences structured in English ?

11. Write an essay on 'Hints on essay-writing'.

12. Write a letter to Railway Company, complaining that your furniture has been damaged in transit, and claiming damages.

13. Imagine that you interviewed the Inspector General of Police after the arrest of a Criminal accused of committing two murders. Write this interview in a form of Newspaper report.

14. How does Public speaking skill help in effective presentation ?

15. Write a Group Discussion on any controversial topic.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

NUTRITION

(2008 Onwards)

Time : 3 Hours | Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Obesity.

2. Micronutrients.

3. Vitamin A.

4. Function of Iron.

5. Diet in Pregnancy.

6. PEM.

7. Functions of Vitamin C.

8. Source of Fat.

Part - B (4 × 10 = 40)

Answer any **four** questions.

9. Discuss Fat Soluble Vitamins.

10. Discuss role of Carbohydrates in Diet.

11. RDA.

12. Functions of Minerals.

13. Discuss BMR.

14. Discuss PEM.

15. Discuss Atherosclerosis.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

VISUAL OPTICS

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Write about purkinge images.

2. Depth of focus.

3. Depth of field.

4. Possible mechanism of accomodation.

5. Accomodation versus Age.

6. Oblique astigmatism.

7. Objective refraction.

8. Contrast sensitivity.

Part - B (4× 10 = 40)

Answer any **four** questions.

9. Retinoscopy

10. Visual Acuity.

11. Myopia.

12. Hypermetropia.

13. Astigmatism.

14. Aphakia.

15. Aberrations.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

CLINICAL REFRACTION—I

(2008 onwards)

Time : 3 Hours | Maximum : 70 Marks

Section - A (5 × 6 = 30)

Answer any **five** questions.

1. Explain the principle of Snellens visual acuity charts.

2. Mention the differences between Static and Dynamic retinoscopy.

3. Explain any one method of measuring amplitude of accommodation.

4. Why is it necessary to check NRA-PRA before prescribing for presbyopes.

5. Write short notes on objective Autorefractor.

6. Write short notes on the principle of retinoscopy.

7. Write short notes on near vision charts.

8. Write short notes on any two types of dynamic retinoscopy.

Section - B (4 × 10 = 40)

Answer any **four** questions.

9. Explain Log MAR charts. Discuss the difference between Snellens and Log MAR.

10. Write the techniques used in retinoscopy in diagnosing astigmatism.

11. Write the principle and usage of JCC.

12. What is cycloplegic refraction. Write in detail on cycloplegics and mydriatics.

13. Explain (a) Astigmatic dial (b) Duochrome (c) Prism dissociation.

14. Explain binocular refraction in detail.

15. Explain (a) Pinhole (b) Stenopaeic slit (c) vertex distance (d) IPD.

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B.Sc., DEGREE EXAMINATION, NOVEMBER 2010

Optometry

OCULAR DISEASES—I

(2008 Onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** Questions.

1. Dry Eye.

2. Dacryocystitis.

3. Explain about keratoplasty.

4. Evisceration.

5. Buphthalmos.

6. Keratoconus.

7. Ectropion and Entropion.

8. Nanophthalmos.

Part - B (4 × 10 = 40)

Answer any **four** Questions.

9. Corneal Ulcers.

10. Cataract.

11. Explain the pathway of aqueous formation and drainage.

12. Explain about POAG.

13. Explain about phacoemulsification.

14. Differentiate scleritis and episcleritis.

15. Explain about the degenerative conditions of conjunctiva.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

OPHTHALMIC INSTRUMENTATION—I

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Section - A (5 × 6 = 30)

Answer any **five** questions.

1. Uses and drawbacks of keratometer.

2. Differentiate Huygens and Ramsden eyepieces.

3. Explain the need and importance of working distance.

4. Explain Snellens fraction, give the difference between Snellen and Bailey lovie chart.

5. Charts for near vision.

6. Additive lens principle.

7. Write about simple microscope.

8. What are the advantages of projection Focimeter ?

Section - B (4 × 10 = 40)

Answer any **four** questions.

9. Explain the different types of keratometer.

10. Explain the objective methods for testing visual acuity.

11. Explain in detail about direct ophthalmoscopes.

12. Types of Retinoscopy and its principle. Explain.

13. Explain refracting units in detail.

14. Explain the optical system in slit lamp.

15. Difference between manual and auto lensometer ?

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

GENERAL AND OCULAR PHARMACOLOGY

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Explain briefly about Newer drug delivery system.

2. Write short notes on NS AIDS.

3. Short notes on β -Blockers.

4. Short notes on preparations and penetration of Ophthalmic drugs.

5. Short notes on the nature and sources of drugs.

6. Topical Anaesthetics used in eye.

7. Short notes on CNS stimulants.

8. Short notes on Ophthalmic diagnostic drugs.

Part - B $(4 \times 10 = 40)$

Answer any **four** questions.

9. Write in detail about routes of drug administration, absorption bio availability distribution and fate of drugs.

10. Define, classify and explain the following categories of drugs with regards to pharmacological action, side effects and therapeutic uses.

(i) General Anaesthetics.

(ii) Anti convulsants.

11. Classification of anti fungal drugs and their use in Ophthalmology.

12. Write short notes on :

(i) Newer Antiglaucoma drugs.

(ii) Newer Antibiotics.

13. Write in detail about uses of steroids and immuno suppressants in the eye.

14. What are the anti allergic drugs used in Ophthalnology ? Add a note on newer arrivals in this category.

15. Define and classify Adverse Drug Reaction (ADR). Explain the manifestation of ADR and a note on treatment of acute drug poisoning.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

CONTACT LENSES—I

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Section - A (5 × 6 = 30)

Answer any **five** questions.

1. Find the spectacle power when CL power is = 7.00 and –7.00D (consider d = 12mm and write to nearest 0.25 D power)

2. What is oxygen transmissibility ? How is it denoted ?

3. List the various parameters which affect soft contact lens design.

4. Write on the assessment of fit of soft lenses.

5. Brief on the history of contact lenses.

6. What is the role of fluorescein in contact lens fitting?

7. Write short notes on over refraction.

8. Classify soft contact lens materials as per FDA.

Section - B $(4 \times 10 = 40)$

Answer any **four** questions.

9. Advantages and disadvantages of Contact lenses over Spectacles.

10. Write in detail on care and maintenance of RGP lenses.

11. Advantages and disadvantages of lathing and moulding methods.

12. Indications and Contra indications of Contact lens.

13. Explain how verification is done on Contact lenses from the laboratory.

14. Advantages and disadvantages of RGP and Soft lenses.

15. Write on the evaluations done preliminary to contact lens fitting.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

BINOCULAR VISION—I

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Part - A (5 × 6 = 30)

Answer any **five** questions.

1. Anisometropia.

2. Convergence.

3. Explain about physiological diplopia.

4. Aniseikonia.

5. ARC.

6. Suppression.

7. Esotropia.

8. Hypotropia.

Part - B (4 × 10 = 40)

Answer any **four** questions.

9. Discuss about Heterophoria.

10. Derive the relationship between Accommodation and convergence.

11. Discuss about the grades of BSV.

12. Discuss in detail about Amblyopia.

13. Write in detail on convergence insufficiency.

14. Discuss about ocular movement.

15. Anatomy of Extra ocular muscles.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

PEDIATRIC OPTOMETRY AND GERIATRIC OPTOMETRY

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Section - A (5 × 6 = 30)

Answer any **five** questions.

1. Write short notes on preschool vision testing.

2. Explain the changes in cornea and lens in old age.

3. Prescribing aspects of hyperopia in pediatrics. Explain briefly.

4. Clinical features and treatment of congenital esotropia.

5. Write on the anomalies of accommodation.

6. Write short notes on Involutional Ectropion and Entropion.

7. Write short notes on treatment of Aphakia.

8. What are the structural anomalies to be looked for while examining the eyelids ?

Section - B (4 × 10 = 40)

Answer any **four** questions.

9. Explain how pediatric history taking is different from the routine History taking.

10. What is Optokinetic Nystagmus ?

11. Write in detail on Age Related mascular degeneration.

12. Write in detail on amblyopia.

13. Explain the different tests to be done for prescribing in presbyopia.

14. Write in detail on Open angle Glaucoma.

15. Write the method and interpretation of cover tests.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Optometry

HOSPITAL PROCEDURES

(2008 onwards)

Time : 3 Hours Maximum : 70 Marks

Section - A (5 × 6 = 30)

Answer any **five** questions.

1. What are the precautions to be taken before removing eye [Enucleation] ?

2. Explain about coding system in Medical records department.

3. How will you explain / counsel the patient before dilatation and refraction ?

4. Explain about the importance of house keeping department in the hospital.

5. Write about legal requirement and importance of medical records.

6. Activities of biochemistry department in eye hospital.

7. Postings in Human resources (HR) department.

8. Write about five ophthalmic instruments and its uses.

Section - B (4 × 10 = 40)

Answer any **four** questions.

9. Explain about Human Resources Department.

10. Explain about :

    (a) Posting and activities of correspondence with patient.

    (b) Assurance of appointment and various methods of cancellation of appointments

11. Different methods of preserving enucleated eye in eye banks Factors affecting the quality of donor cornea ?

12. Role of optomery department in eye hospital and its activities.

13. Explain about reception services.

14. Explain about Medical records department.

15. What are all the pretest to be done before cataract surgery and how will you counsel the patient before surgery ?

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